

CBSE Class 10 Physics: Electricity

Complete Concept Notes, Formulas, Solved Examples & Exam Practice Questions

Comprehensive Study Guide • Aligned with NCERT Curriculum • Board Exam Preparation

1. Electric Charge, Current, and Potential

1.1 Electric Charge (Q)

Electric charge is an intrinsic property of matter. It exists in two types: positive (protons) and negative (electrons). Like charges repel, and opposite charges attract.

- **SI Unit:** Coulomb (C)
- **Charge on an Electron:** $e = 1.6 \times 10^{-19} C$
- **Quantization of Charge:** Total charge $Q = n \cdot e$, where n is an integer (number of electrons).

Formula: $Q = n \cdot e$ | $1 C \approx 6.25 \times 10^{18}$ electrons.

1.2 Electric Current (I)

Electric current is defined as the rate of flow of electric charge through any cross-section of a conductor.

$$I = Q / t \quad \&\text{implies}; \quad Q = I \cdot t$$

- **SI Unit:** Ampere (A). Named after André-Marie Ampère.
- **1 Ampere:** Current produced when 1 Coulomb of charge flows through a conductor in 1 second ($1 A = 1 C / 1 s$).
- **Smaller Units:** $1 mA = 10^{-3} A$, $1 \mu A = 10^{-6} A$.
- **Measurement:** Measured using an **Ammeter**, which is always connected in *series* in a circuit (low resistance).
- **Direction:** Conventionally taken opposite to the flow of negative electrons (from positive terminal to negative terminal).

1.3 Electric Potential & Potential Difference (V)

Electric Potential Difference between two points in an electric circuit is the amount of work done in moving a unit positive charge from one point to another.

$$V = W / Q$$

- **SI Unit:** Volt (V). Named after Alessandro Volta.
- **1 Volt:** Potential difference between two points when 1 Joule of work is done to move a charge of 1 Coulomb ($1 V = 1 J / 1 C$).
- **Measurement:** Measured using a **Voltmeter**, connected in *parallel* across the component (high resistance).

Solved Example 1.1

Question: How many electrons constitute a charge of 1 C ?

SOLUTION:

Given: $Q = 1\text{ C}$, $e = 1.6 \times 10^{-19}\text{ C}$

Using $Q = n \cdot e$ &implies; $n = Q / e = 1 / (1.6 \times 10^{-19}) = 6.25 \times 10^{18}$ electrons.

Solved Example 1.2

Question: A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit.

SOLUTION:

Given: $I = 0.5\text{ A}$, $t = 10\text{ min} = 10 \times 60 = 600\text{ s}$

Using $Q = I \times t = 0.5\text{ A} \times 600\text{ s} = 300\text{ C}$.

2. Ohm's Law and Resistance

2.1 Ohm's Law

Ohm's Law states that *the electric current flowing through a metallic conductor is directly proportional to the potential difference across its ends, provided its temperature and physical conditions remain constant.*

$$V \propto I \text{ \&implies; } V = I \cdot R$$

2.2 Resistance (R)

Resistance is the property of a conductor to resist/oppose the flow of electric charges through it.

- **SI Unit:** Ohm (Ω).
- **1 Ohm:** If potential difference across a conductor is 1 V and current flowing is 1 A , resistance is $1\ \Omega$ ($1\ \Omega = 1\text{ V} / 1\text{ A}$).

2.3 Factors Affecting Resistance

The resistance of a uniform metallic conductor depends on:

1. **Length (l):** $R \propto l$ (Directly proportional)
2. **Area of Cross-Section (A):** $R \propto 1/A$ (Inversely proportional)
3. **Nature of Material**
4. **Temperature:** Resistance increases with an increase in temperature for metallic conductors.

$$R = \rho \cdot (l / A)$$

where ρ (rho) is the **Resistivity** (or Specific Resistance) of the material.

2.4 Resistivity (ρ)

- **SI Unit:** Ohm-meter ($\Omega\cdot\text{m}$).
- Characteristic property of a material; does **not** depend on length or thickness/area of the conductor.

- Metals & Alloys: Low resistivity ($10^{-8} \Omega \cdot m$ to $10^{-6} \Omega \cdot m$) → Good conductors.
- Insulators: High resistivity ($10^{12} \Omega \cdot m$ to $10^{17} \Omega \cdot m$).
- Alloys (e.g., Nichrome, Constantan) have higher resistivity than constituent metals and do not oxidize (burn) easily at high temperatures. Hence used in heating elements.

Solved Example 2.1

Question: Resistance of a metal wire of length 1 m is 26Ω at 20°C . If the diameter of the wire is 0.3 mm , what will be the resistivity of the metal?

SOLUTION:

Given: $R = 26 \Omega$, $l = 1 \text{ m}$, diameter $d = 0.3 \text{ mm} = 3 \times 10^{-4} \text{ m}$.

Radius $r = d / 2 = 1.5 \times 10^{-4} \text{ m}$.

Area $A = \pi r^2 = 3.14 \times (1.5 \times 10^{-4})^2 \approx 7.065 \times 10^{-8} \text{ m}^2$.

Since $R = \rho l / A$ $\implies \rho = RA / l$:

$\rho = (26 \times 7.065 \times 10^{-8}) / 1 \approx 1.84 \times 10^{-6} \Omega \cdot m$.

3. Combination of Resistors

3.1 Resistors in Series

When resistors are connected end-to-end sequentially, they are in series.

- **Current (I):** Remains the *same* through all resistors.
- **Potential Difference (V):** Total voltage equals sum of potential drops across individual resistors ($V = V_1 + V_2 + V_3$).
- **Equivalent Resistance (R_s):**

$$R_s = R_1 + R_2 + R_3 + \dots + R_n$$

3.2 Resistors in Parallel

When resistors are connected together between two common nodes, they are in parallel.

- **Potential Difference (V):** Same across each resistor.
- **Current (I):** Total current equals sum of currents through individual branches ($I = I_1 + I_2 + I_3$).
- **Equivalent Resistance (R_p):**

$$1 / R_p = 1 / R_1 + 1 / R_2 + 1 / R_3 + \dots + 1 / R_n$$

Parameter	Series Combination	Parallel Combination
Current	Same through all resistors	Divides among branches ($I = \sum I_i$)
Voltage	Divides across resistors ($V = \sum V_i$)	Same across all branches
Equivalent R	$R_s = R_1 + R_2 + \dots$ (Maximum)	$1/R_p = 1/R_1 + 1/R_2 + \dots$ (Minimum)
Failure Impact	If one breaks, circuit open (all off)	If one breaks, others keep working

Parameter	Series Combination	Parallel Combination
Application	Decorative light strings	Domestic household wiring

Solved Example 3.1

Question: Three resistors $5\ \Omega$, $10\ \Omega$, and $30\ \Omega$ are connected in parallel across a $12\ \text{V}$ battery. Calculate: (a) equivalent resistance, (b) total current in circuit.

SOLUTION:

(a) $1/R_p = 1/5 + 1/10 + 1/30 = (6 + 3 + 1)/30 = 10/30 = 1/3$ & implies; $R_p = 3\ \Omega$.

(b) Total current $I = V/R_p = 12\ \text{V}/3\ \Omega = 4\ \text{A}$.

4. Heating Effect of Electric Current & Power

4.1 Joule's Law of Heating

When an electric current flows through a purely resistive conductor, electrical energy is converted entirely into heat energy.

According to Joule's Law, heat produced (H) in a conductor is:

$$H = I^2 \cdot R \cdot t$$

- Directly proportional to the square of current (I^2).
- Directly proportional to resistance (R).
- Directly proportional to time (t) for which current flows.

4.2 Electric Power (P)

Electric power is the rate at which electrical energy is consumed or dissipated in a circuit.

$$P = W/t = V \cdot I = I^2 \cdot R = V^2/R$$

- **SI Unit:** Watt (W). $1\ \text{W} = 1\ \text{J/s} = 1\ \text{V} \cdot 1\ \text{A}$.
- **Commercial Unit of Energy:** Kilowatt-hour (kWh) or "Unit".
- $1\ \text{kWh} = 1000\ \text{W} \times 3600\ \text{s} = 3.6 \times 10^6\ \text{J (Joules)}$.

Solved Example 4.1

Question: An electric refrigerator rated $400\ \text{W}$ operates 8 hours/day. What is the cost of energy to operate it for 30 days at ₹ 3.00 per kWh ?

SOLUTION:

Total energy consumed = $\text{Power} \times \text{Time} = 400\ \text{W} \times (8\ \text{h/day} \times 30\ \text{days}) = 400\ \text{W} \times 240\ \text{h} = 96,000\ \text{Wh} = 96\ \text{kWh}$.

Total Cost = $96\ \text{kWh} \times ₹\ 3.00/\text{kWh} = ₹\ 288.00$.

5. Important Practice Questions & Solutions

Question 1 (Short Answer - 2 Marks)

Why are copper and aluminum wires usually employed for electricity transmission, while tungsten is used almost exclusively for filament of electric lamps?

SOLUTION:

- **Copper & Aluminum:** Have very low electrical resistivity, making them excellent conductors. They transmit electrical energy with minimal heat loss (I^2Rt).
- **Tungsten:** Has a very high melting point (3380°C) and high resistivity. It retains heat energy to become white-hot and emit light without melting or oxidizing at high temperatures.

Question 2 (Numerical - 3 Marks)

A wire of given material having length l and area of cross-section A has a resistance of $4\ \Omega$. What would be the resistance of another wire of the same material having length $l/2$ and area of cross-section $2A$?

SOLUTION:

For First Wire: $R_1 = \rho (l / A) = 4\ \Omega$.

For Second Wire: $l' = l/2$, $A' = 2A$.

$$R_2 = \rho (l' / A') = \rho ((l/2) / (2A)) = \rho (l / 4A) = (1/4) \cdot [\rho(l/A)] = (1/4) \cdot R_1.$$

$$R_2 = 4\ \Omega / 4 = 1\ \Omega.$$

Question 3 (Numerical - 3 Marks)

Two electric bulbs rated $220\ \text{V}$, $100\ \text{W}$ and $220\ \text{V}$, $60\ \text{W}$ are connected in parallel to an electric mains supply. What current is drawn from the line if the supply voltage is $220\ \text{V}$?

SOLUTION:

Since both bulbs are connected in parallel across $220\ \text{V}$:

$$\text{Current for Bulb 1: } I_1 = P_1 / V = 100\ \text{W} / 220\ \text{V} = 5 / 11\ \text{A}.$$

$$\text{Current for Bulb 2: } I_2 = P_2 / V = 60\ \text{W} / 220\ \text{V} = 3 / 11\ \text{A}.$$

$$\text{Total Current drawn: } I = I_1 + I_2 = 5/11 + 3/11 = 8/11\ \text{A} \approx 0.727\ \text{A}.$$

Question 4 (Circuit Analysis - 5 Marks)

In the given circuit diagram setup: Three resistors of $R_1 = 10\ \Omega$, $R_2 = 20\ \Omega$, and $R_3 = 30\ \Omega$ are connected in parallel, and this parallel combination is connected in series with a resistor $R_4 = 6\ \Omega$ across a battery of $12\ \text{V}$.

Calculate: (a) Equivalent resistance of parallel group, (b) Total circuit resistance, (c) Total circuit current, (d) Voltage drop across R_4 .

SOLUTION:

(a) Parallel Group (R_p):

$$1/R_p = 1/10 + 1/20 + 1/30 = (6 + 3 + 2)/60 = 11/60 \text{ \&implies; } R_p = 60/11 \, \Omega \approx 5.45 \, \Omega.$$

(b) Total Resistance (R_{total}):

$$R_{total} = R_p + R_4 = 5.45 \, \Omega + 6 \, \Omega = 11.45 \, \Omega \text{ (or } 60/11 + 6 = 126/11 \, \Omega).$$

(c) Total Current (I):

$$I = V / R_{total} = 12 \, \text{V} / (126/11 \, \Omega) = (12 \times 11) / 126 = 132 / 126 \approx 1.047 \, \text{A}.$$

(d) Voltage drop across R_4 (V_4):

$$V_4 = I \times R_4 = 1.047 \, \text{A} \times 6 \, \Omega \approx 6.28 \, \text{V}.$$

Question 5 (Conceptual - 2 Marks)

Why does the cord of an electric heater not glow while the heating element does?

SOLUTION:

The cord of an electric heater is made of copper which has extremely low resistance (R). Therefore, according to Joule's Law ($H = I^2 R t$), negligible heat is generated in the cord.

Conversely, the heating element is made of high-resistivity alloy (like Nichrome), which produces a very high amount of heat energy for the same current, causing it to become red-hot and glow.